



Assessment of two ensemble-based rapid-update data assimilation and short-range numerical weather prediction systems for Argentina

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Abstract

PREVENIR—Forecast and Warning of Flash Flood Events is an Argentina-Japan cooperation project for five years from 2022 that aims to develop an early warning system for heavy rainfall and urban floods. The current operational numerical weather prediction system of the Argentine National Meteorological Service consists of deterministic and probabilistic forecasts at 4 km resolution. Since PREVENIR aims at more accurate and timely precipitation forecast, we developed a 2 km resolution 5-min update data assimilation system with the Local Ensemble Transform Kalman Filter (LETKF) that assimilates observation data from an automated weather station network and C-band Doppler weather radars. Independent LETKF systems have been coupled with the regional models' Weather Research and Forecasting (WRF) and Scalable Computing for Advanced Library and Environment (SCALE), for two target basins: a mountainous region of the Suquia Villa Paez in Córdoba Province and a flat region of the Sarandí Santo Domingo in Buenos Aires. This study investigates the performance of the two systems in two extreme rain cases. We show that the forecasted precipitation benefits from the high-resolution rapid-update process, which improves its location and intensity with respect to coarser forecasting resources available in the region and also non-data assimilation systems. The results suggest that the prototypes could offer several advantages for hydrological applications, representing a potentially valuable tool for the region.

Keywords PREVENIR · Precipitation forecast · Rapid-update data assimilation · LETKF · SCALE · WRF

Abbreviations

AWS	Automatic weather station	GFS	Global forecasting system
DA	Data assimilation	LETKF	Local ensemble transform Kalman filter
FSS	Fractions skill score	Lin	Lin microphysics scheme
		MYJ	Mellor-Yamada-Janjic planetary boundary layer

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	scheme
MYNN	Mellor-Yamada Nakanishi-Niino parameterization
MODE	Method for object-based diagnostic evaluation
Noah-MP	Noah land surface model with multiparameterizations
PREVENIR	Forecast and warning of flash flood events research cooperation project
R	Convolution radius (used for MODE)
RELAMPAGO	Remote sensing of electrification, lightning, and mesoscale/microscale processes with adaptive ground observations
RMSE	Root mean square error
RQPE	Radar quantitative precipitation estimation
RRTMG	Rapid radiative transfer model parameterization
SAP.SMN	Data assimilation and numerical weather forecasting system of the Argentine national meteorological service
SCALE	Scalable computing for advanced library and environment
SH	Shin-Hong planetary boundary layer scheme
SMN	Argentine national meteorological service
SSD	Sarandí-Santo Domingo Creek basin
SVP	Suquía-Villa Páez river basin
TH	Convolution threshold (used for MODE)
WRF	Weather research and forecasting model
WSM6	WRF single-moment 6-class microphysics scheme
YSU	Yonsei University planetary boundary layer scheme

Table 1 Operational hourly-cycling convection-permitting systems used by meteorological centers worldwide

Meteorological center	Data assimilation system	Horizontal resolution	References
Met office	4D-Var	1.5 km	Milan et al (2020)
National oceanic and atmospheric administration	hybrid En-3D-Var	3 km	Dowell et al. (2022); Yussouf and Knopfmeier (2019); Wheatley et al. (2015); Wilson et al. (2023)
Meteo-France	3D-Var	1.3 km	Brousseau et al (2016)
Deutscher Wetterdienst	nudging-LETKF	2.2 km	Schraff et al. (2016)
Japan Meteorological Agency	3D-Var	2 km	Aranami et al. (2015); Ikuta et al (2021)
Argentine National Meteorological Service	LETKF	4 km	Matsudo et al. (2025)

1 Introduction

Urban floods are the deadliest hazard associated with deep moist convection worldwide, affecting a large part of the population since more than half the world's population lives in urban areas (WHO 2015; Markowski and Richardson 2010). In the largest and more densely populated regions, flooding is a frequent phenomenon, with several events occurring yearly and thousands of people being directly affected, particularly those living in vulnerable neighborhoods. One of the most important ways to mitigate the impact of urban flash floods is an effective and timely early-warning system so that both the decision-makers and the residents can be aware of the risks and take action to protect themselves and their property.

Rasmussen et al. (2014) showed that Central and Northern Argentina is affected by flash floods associated with heavy convective precipitation every year. In addition, over Southeast South America, a positive trend in extreme precipitation occurrence was detected during the last decades both in monthly and daily scales, mainly for warm seasons (e.g., Haylock et al. 2006; Penalba and Robledo 2010; Barros et al. 2013). Moreover, future projections generally agree in that the frequency and intensity of precipitation extremes will increase across most land regions (Li et al. 2021). Consequently, forecasting extreme precipitation events will become even more challenging and critical in the forthcoming future.

Several studies showed the significant impact of regional data assimilation systems on precipitation forecasts, particularly those working at convection permitting scales with radar observations (e.g., Gustafsson 2018). Dowell et al. (2022) and Hu et al. (2023) documented some of the operational hourly-cycling convection-permitting systems used by meteorological centers worldwide, which are summarized in Table 1.

Particularly, the operational Data Assimilation and Numerical Weather Forecasting System of the Argentine National Meteorological Service (known by its Spanish acronym SAP.SMN) consists of warm-start deterministic and probabilistic forecasts, initialized four times a day with 4-km resolution (Matsudo et al. 2022), using the Weather Research and Forecasting model (WRF; Skamarock et al. 2008). The warm-start is achieved through an hourly-cycling Local Ensemble Transform Kalman Filter (LETKF; Hunt et al. 2007; Miyoshi and Kunii (2012);) data assimilation system, which has been used for the probabilistic forecasts initializations since December 2024 (Matsudo et al. 2025) and for the deterministic ones since November 2025.

Although the experiences at different meteorological centers around the world have been encouraging, these convection-permitting systems are not exempt from scientific

challenges such as, for example, fast error growth and complex observation operators (e.g., Yano et al. 2018). In this context, the PREVENIR—Forecast and Warning of Flash Flood Events research cooperation project between Argentina and Japan began in 2022 as a 5-year project, to develop an urban flash flood early warning system for two pilot basins among the most populated regions in Argentina, through an interdisciplinary workflow organized in seven work packages. One of the PREVENIR's key components is a big data assimilation and numerical weather prediction system capable of producing frequently updated operational precipitation forecasts that can provide valuable information for end decision-makers and hydrological prediction systems (<https://sites.google.com/view/prevenir-en/home>, accessed October 31st, 2025).

The objective of this work is to explore the performance of two prototypes for the PREVENIR system coupling the LETKF, independently, with two different regional models: WRF and Scalable Computing for Advanced Library and Environment (SCALE; Nishizawa et al. 2015), taking into account the feasibility of a subsequent transition from research to operations. Each prototype is based on a convection-permitting (2-km resolution) frequently-updated ensemble-based data assimilation and short-range forecast system, incorporating observations from an automated surface weather station network and C-band Doppler weather radars. We investigate the predictability of two intense precipitation events associated with urban flash floods, using grid-point estimation of error and uncertainty together with an object-based approach, to evaluate the high-frequency precipitation forecasts.

The article is structured as follows. Section "Experimental design" describes the experimental design, including a description of the pilot basins and test cases, the characteristics of the models, the assimilated observations, and the ensemble data assimilation and forecast system configuration. Sections 3 and 4 present the results obtained for both cases, focusing on the analysis and short-range forecast verification, respectively. Finally, Sect. 5 summarizes the main findings.

2 Experimental design

2.1 Pilot basins and test cases

The two pilot basins of the PREVENIR project were selected based on the different features each of these basins have, and the vulnerability of the population living in them which augments the urgency of having increasingly accurate precipitation forecasts to fulfill the community's needs. The Sarandí-Santo Domingo Creek Basin (SSD) in Buenos

Aires Province plains and the Suquía-Villa Páez River Basin (SVP) in the mountainous region of Sierras de Córdoba, are among the most populated regions in Argentina (Fig. 1). Although the SSD basin is relatively small, with an area of 240 km², the population is very vulnerable due to significant urban growth over the last decades and inappropriate infrastructure of, for example, the drainage systems (Re et al. 2022). Primarily, the population settled without basic services, including a stormwater drainage system, in a naturally flood-prone wetland area. Regarding the SVP basin, it covers an area of 2500 km² with steep mountainous terrain, making it a fast-response basin prone to flash floods. In addition, it presents a dam acting as a regulator, containing floods upstream of Córdoba city, and an extended area of rural conditions downstream together with a substantial urban area (López et al. 2023).

The SVP case took place at the end of December 13th, 2018, when a broad upper-level trough moved towards central Argentina, contributing to the deepening of the Northwestern Argentinian Low (Seluchi et al. 2003) and the advance of a surface cold front towards central Argentina. In an environment characterized by large CAPE and deep-layer shear, long-lived supercells developed during the afternoon and evening of the 13th over the south and central SVP domain. Explosive convective development and rapid upscale growth into a linear Mesoscale Convective System occurred along the leading edge of an intense cold pool emerging from those cells that interacted with the central and northern Sierras de Córdoba (Fig. 2a, c). Numerous severe weather reports were collected, including urban flooding in Córdoba City. This event occurred during an Intensive Operation Period mission from the Remote sensing of Electrification, Lightning, And Mesoscale/microscale Processes with Adaptive Ground Observations (RELAMPAGO) field campaign (Nesbitt et al. 2021), and has been studied from different perspectives by Schumacher et al. (2021), Rocque and Rasmussen (2022), Rocque et al. (2024), Sasaki et al. (2025).

The SSD case started on October 10th, 2019, when low-level confluence over central Argentina led to strong frontogenesis West–East oriented extending from the Andes to the La Plata River. Several storms initiated along this boundary, and propagated in the same easterly direction, driven by the prevailing westerly winds (Fig. 2b, d). As a result of this pattern, the SSD basin was continuously affected by storms for several hours from the early morning of October 11th, leading to large precipitation totals (up to 80 mm) and urban floods, with streets and avenues inundated, and streams and drainage systems overflowing.

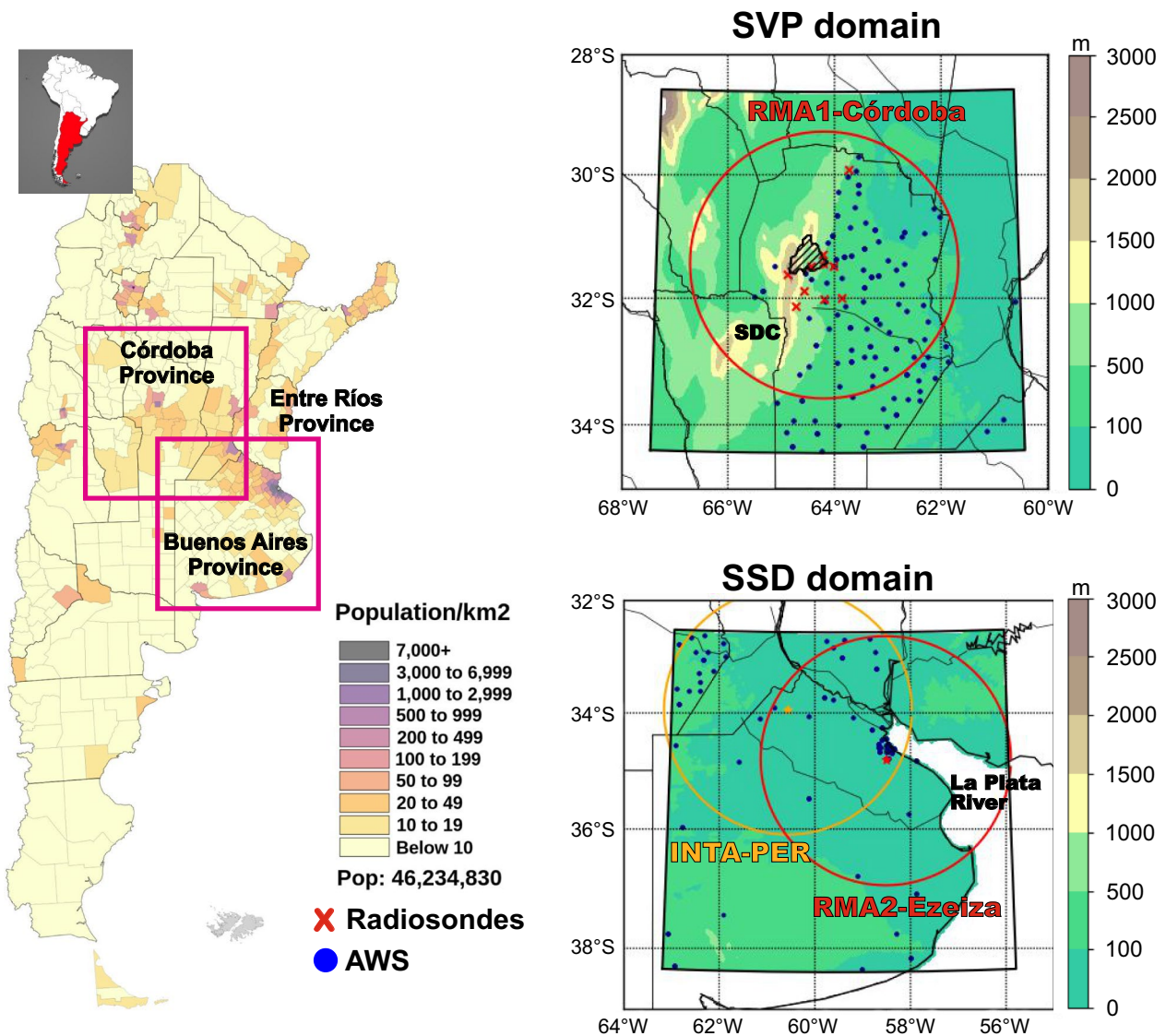


Fig. 1 Computational domains for SVP and SSD cases, indicating SCALE model topography (shaded; m) and assimilated observation sources location, AWS (blue dots), and radars (RMA1-Córdoba and RMA2-Ezeiza in red circles, INTA-PER in yellow). SVP basin is depicted with a hatched polygon, the location of radiosondes used for validation is indicated by red crosses, and the mountainous region

of Sierras de Córdoba is specified by SDC. SSD basin is located at approximately 34.8°S–58.3°W. The La Plata River and the provinces of Córdoba, Entre Ríos, and Buenos Aires are shown in the maps to facilitate text interpretation. Argentina's population density according to the 2022 Census Demographic Data Map Viewer together with its location in South America is shown for reference

2.2 Numerical weather prediction models

Numerical experiments were conducted using two models; the WRF model with its Advanced Research WRF dynamical solver version 4.0 (Skamarock et al. 2019; referred to as WRF from now on) and the SCALE-RM version 5.4.5 (Nishizawa et al. 2015; referred to as SCALE from now on). It is worth mentioning that, although similar arrangements were selected across the models to ensure a consistent basis for comparison, the primary criterion was to obtain the most suitable configuration for each model taking into account findings from previous studies. This decision reflects the

main objective of this work, namely, to explore the performance of PREVENIR prototypes under conditions that maximize their respective capabilities and to assess the feasibility of a subsequent transition from research to operations.

The WRF model has been extensively used in multiple studies in the Argentine region (e.g., Ruiz et al. 2010; Alvarez Imaz et al. 2021; Casaretto et al. 2022), and it is the operational regional model used within the Argentine SMN (Matsudo et al. 2022). A multiphysics approach was introduced for cloud microphysics and planetary boundary layer parameterizations to produce different ensemble members.

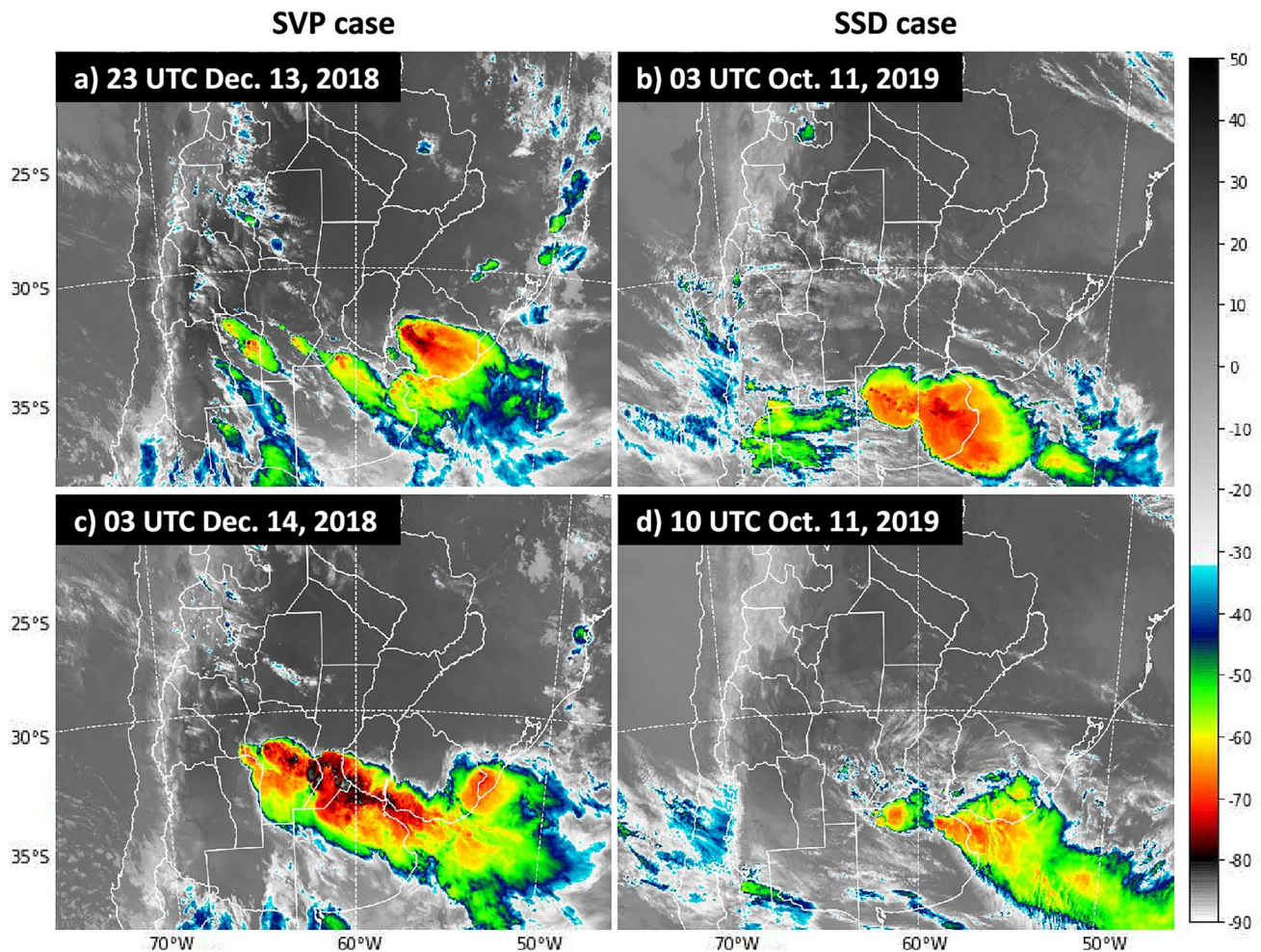


Fig. 2 Brightness temperature (°C) from channel 13 ABI sensor on board of GOES-16 for the SVP case (left column) and SSD case (right column). **a** 23 UTC on December 13th, 2018, **c** 03 UTC on December 14th, 2018, **b** 03 UTC on October 11th, 2019, and **d** 10 UTC on October 11th, 2019

The multiphysics model integrations are expected to better represent the model uncertainty associated with the parameterization of unresolved physical processes. The planetary boundary layer schemes used were: Mellor-Yamada-Janjic (MYJ; Janjic 1994), Yonsei University (YSU; Hong et al. 2006), and Shin-Hong (SH; Shin and Hong 2015). Cloud microphysics processes were parameterized with the WRF single-moment 6-class microphysics scheme (WSM6) (Hong and Lim 2006) and Lin (Lin et al. 1983). For both shortwave and longwave radiation, the Rapid Radiative Transfer Model (RRTMG) parameterization was used (Iacono et al. 2008), while the Noah land surface model with Multiparameterizations (Noah-MP) was selected for the land surface processes (Niu et al. 2011).

The SCALE model was developed by the RIKEN Center for Computational Science with co-design by specialists in weather and climate modeling and computational science, aiming for efficient computation in various high-performance computing environments. The model has been

used for advanced meteorological studies using the supercomputer K, the previous generation of Japanese flagship supercomputer, and the current one, the supercomputer Fugaku (Miyoshi et al. 2016, 2023), and it has also been evaluated in Argentina by Maldonado (2023) obtaining encouraging results. The parameterization schemes used in the experiments include a single-moment 6-category cloud microphysics parameterization (Tomita 2008), a Smagorinsky-type subgrid-scale turbulence parameterization (Smagorinsky 1963), the level 2.5 closure of Mellor-Yamada Nakanishi-Niino (MYNN) type boundary layer parameterization (Nakanishi et al. 2004), and the Model Simulation radiation Transfer code version X parameterization (mstrnX; Sekiguchi and Nakajima 2008).

Both models were applied in two different domains of 640×640 km, one encompassing the SVP and the other encompassing the SSD basins (c.f., Fig. 1), with 2 km horizontal grid spacing and using a Lambert projection. The number of vertical levels was set to 60, with the top defined

at approximately 25 km for SCALE, and 10 hPa (~30 km) for WRF. No cumulus parameterization was used for either of the models. Table 2 shows a summary of the model's configuration.

It should be noted that some configuration choices may influence the analyses and forecasts, such as the different treatment of the land surface or the definition of the ensemble. These aspects will therefore be taken into account when discussing and interpreting the results.

2.3 Assimilated observations

The assimilated observations include data from automatic surface weather stations (AWS) and C-band weather radars, both provided by the SMN. The observation's spatial distribution is presented in Fig. 1.

AWS data consists of 2-m temperature, 2-m relative humidity, surface pressure, and 10-m zonal and meridional wind (c.f., blue dots in Fig. 1). These observations belong to public and private surface networks, such as Córdoba and Entre Ríos province's board of trade, the National Institute for Agricultural Technology, and the University of San Luis. Most AWS provided data every 10 min, however, a few stations provided data at higher frequencies (up to every 1 min). Considering the 5-min frequency DA scheme used, which is described in Sect. "Ensemble data assimilation and forecast system", high-frequency observations were superobbed in time to a 5-min frequency. Observational errors of 2-m temperature, 2-m relative humidity, surface pressure, and 10-m zonal and meridional wind were set to 2 K, 10%, 1.0 hPa, and 1.4 m/s, respectively, following the recommendations in the WRFDA package (Barker et al. 2012) and previous implementations in the region (e.g. Dillon et al. 2021; Casaretto et al. 2023).

C-band radar data consists of reflectivity and Doppler velocity from the RMA1-Córdoba radar for the SVP case (c.f., red circle in Fig. 1), and RMA2-Ezeiza and INTAPER radars for the SSD case (c.f., red and yellow circles in Fig. 1, respectively). RMA radars belong to the newly developed Argentine C-band Doppler dual-polarization weather radar network (de Elía et al. 2017), while INTAPER is a C-band Doppler single-polarization radar owned

by the National Institute for Agricultural Technology. A pre-processing, consisting of a quality control step followed by a spatiotemporal superobbing, was applied to each radar in its original geometry.

The quality control applied to reflectivity data is similar to the one used at SMN for operations (Arruti et al. 2021). It removes areas strongly affected by attenuation, non-meteorological echoes related to clutter, speckle, anomalous propagation, interference, complex terrain, and second-trip echoes. For Doppler velocity, the SMN experimental (not yet operational) version of the quality control was applied to correct aliasing and smooth the wind field by removing pixels that are not spatially coherent in azimuth (Maldonado 2023).

The spatiotemporal superobbing was employed to convert the radar data from its original resolution (i.e., 500 m in range, 1 degree in azimuth, and several antenna elevations) to a 5-min frequency consistent with the DA frequency (see Sect. "Ensemble data assimilation and forecast system"), and a spatial resolution of 4 km horizontally and 1 km vertically. This procedure involves constructing a regular three-dimensional grid centered on the radar site with the defined resolution. Then, for each grid cell, all radar pixels whose latitude, longitude and elevation fall within the cell boundaries are identified and their values are averaged to produce a single super-observation. To ensure statistical representativeness and reduce sampling noise, a minimum threshold of eight radar observations per grid cell is required; grid cells with fewer than eight observations are discarded. The selection of a 4 km resolution responds to preliminary experiments where using a superobbing horizontal resolution matching the model grid resolution of 2 km, showed a detrimental impact on the analysis, producing a smaller reflectivity spread and some imbalances in the vertical velocity field with noisier analysis increments (not shown).

Observational error was set to 5 dBZ and 2 m/s for superobbed reflectivity and Doppler velocity, respectively, following previous studies in South America (e.g., Dillon et al. 2021; Maldonado 2023). Additionally, the assimilation of clear-air observations (i.e., reflectivity observations lower than 10 dBZ) to suppress spurious convection within the computational domain was considered (Aksoy et al. 2009).

2.4 Ensemble data assimilation and forecast system

Two data assimilation systems based on the LETKF method (Hunt et al. 2007) were used in this work: the WRF-LETKF (Miyoshi and Kunii 2012) and the SCALE-LETKF (Lien et al. 2017). Both systems have already been developed and used for various applications at SMN and RIKEN, respectively (e.g., Dillon et al. 2016, 2021; Maldonado et al. 2020,

Table 2 Summary of SCALE and WRF configuration used for the experiments. For more details please refer to the text

	SCALE	WRF
Microphysics	1-moment Tomita	1-moment WSM6/Lin
Planetary boundary layer	MYNN level 2.5	MYJ/YSU/SH
Radiation	mstrnX	RRTMG
Land surface	bucket	Noah-MP
Vertical levels	60, top at ~25 km	60, top at 10 hPa (~30 km)
Horizontal resolution	2 km (320 × 320 grid points)	

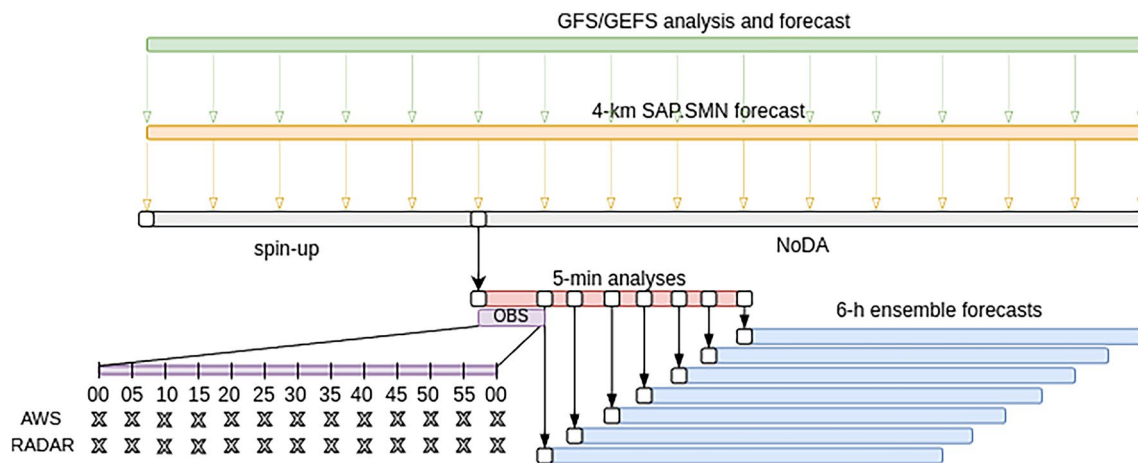


Fig. 3 Simulations flowchart, including the spin-up and NoDA free ensemble forecasts (gray bars), 5-min analyses (red bar) with AWS, and radar observation distribution over a 1-h period (violet bar), 6-h

extended ensemble forecasts initialized by high-resolution analyses (blue bars), and initial and boundary conditions data provided by the 4-km SAP.SMN ensemble forecast (yellow bar)

Table 3 Period for each simulation type (c.f., Fig. 3) for SVP and SSD cases

CASE	spin-up forecast	NoDA forecast	5-min analyses	# of cycles	6-h ensemble forecasts	
	Period	Period	Period		Period	Frequency
SVP (2018)	18–23 UTC 13 Dec	00–09 UTC 14 Dec	23 UTC 13 Dec 03 UTC 14 Dec	48	00–03 UTC 14 Dec	30 min
SSD (2019)	21 UTC 10 Oct 03 UTC 11 Oct	04–16 UTC 11 Oct	03–10 UTC 11 Oct	84	04–10 UTC 11 Oct	30 min

2021; Amemiya et al. 2020; Taylor et al. 2021; Casaretto et al. 2023).

The data assimilation system uses 40 ensemble members, while the extended 6-h ensemble forecasts are initiated from the first 20 analysis ensemble members. A flowchart describing the simulations is depicted in Fig. 3, and the specific period of each simulation is shown in Table 3. Before starting the DA (c.f., red bar in Fig. 3), a 6-h spin-up free model integration is conducted for each model following previous works such as Maldonado et al. (2020), which should be sufficient for the development of finescale structures according to Skamarock (2004). For the initial and boundary conditions for both models, a cold-start configuration from the 4-km SAP.SMN forecasts are used (c.f., yellow bar in Fig. 3). These regional forecasts consist of a cold-start deterministic and 20-member ensemble WRF run, which uses as initial and boundary conditions the Global Forecasting System (GFS) and Global Ensemble Forecasting System, respectively (green bar Fig. 3; Zhou et al. 2017). To construct the 40 initial and boundary conditions for the WRF and SCALE DA cyclings, the perturbations of the SAP.SMN's 20 ensemble members are calculated and added/subtracted to the deterministic SAP.SMN run, obtaining 40 different state conditions. For both cases, the 18 UTC initialization from SAP.SMN was used, although the spin-up integration for the SVP case started at 18 UTC and for the SSD case at 21 UTC (c.f., Table 3).

Observations are assimilated every 5 min and the data collected during the cycle is assimilated as an instantaneous value at the assimilation times (i.e., a 5-min backward assimilation window is used, c.f., violet bar in Fig. 3). We acknowledge that such a high-frequency cycling strategy may introduce dynamical imbalances and that the resulting forecasts do not necessarily outperform those produced with lower-frequency cycling. Previous studies have reported mixed results in this regard (e.g., Lin et al. 2021; Liu et al. 2021; Yang and Wang 2023; Alfie 2026; Amemiya and Miyoshi 2026). Nevertheless, given that the primary objective of the proposed systems is to provide high-frequency precipitation forecasts for hydrological applications, a 5-min cycling interval was selected in order to represent convective processes as accurately as possible. Moreover, spatial kinetic energy spectra were calculated for the ensemble analyses following Bierdel et al. (2012) for each case. It was found that the behaviour of the systems is consistent and similar between them, also reproducing the -5/3 spectral slope for the horizontal velocity components on the meso-scale (not shown).

In both WRF-LETKF and SCALE-LETKF systems, observation localization is used with an approximate Gaussian function proposed by Gaspari and Cohn (1999). Localization length scales of 4 km in the horizontal and 2 km in the vertical are used for radar reflectivity and Doppler velocity following the sensitivity experiments from

Maldonado et al. (2020). For AWS data, a horizontal length scale of 10 km is used and a 0.4 scale in log-pressure units is considered in the vertical, taking into account previous studies and the model's resolution of the present work (Dillon et al. 2021). For spread control, multiplicative inflation with a factor of 1.1 and relaxation-to-prior-perturbation (Zhang et al. 2004) with a factor of 0.8 are considered (Honda et al. 2022). The gross-error check of the observation is applied to reject observations that are too far from the corresponding first guess. The threshold value is 7 times the observation error standard deviation for radar reflectivity and 5 times the observation error standard deviation for Doppler wind and AWS data, in order to discard those with large departures between the observed and the simulated values. The three C-band radars are assimilated independently. One drawback of the reflectivity assimilation is that SCALE limits the amount of observations that can impact a given gridpoint while WRF does not (see Sect. "Verification against AWS and radar observations"). These differences may influence the analyses produced by the data assimilation, and determining which strategy performs better for each system should be addressed in future research.

One hour after initializing the 5-min analysis cycling, 6-h extended forecasts were conducted every 30 min (c.f., blue bars in Fig. 3). A NoDA forecast was also run to provide a basis to assess the impact of DA in short-range ensemble forecasts (Table 3).

2.5 Metrics used for verification

For grid point comparisons, both root mean square error (RMSE) and bias statistics were calculated for different variables. In addition, the probability Fractions Skill Scores (pFSS) of surface precipitation were calculated using the Roberts and Lean (2008) modified version as in Maldonado et al. (2021), to take into account the information provided by all the ensemble members. The Method for Object-based Diagnostic Evaluation tool (MODE; Bullock et al. 2016) was used to perform a feature-based verification of 6-h accumulated precipitation forecasts.

The observational data used as reference was the Radar Quantitative Precipitation Estimation (RQPE), obtained for RMA1 and RMA2. The calculation was performed at a constant elevation of 0.9° , using an R(KDP) relationship obtained from disdrometers deployed within the radar's range. Non-meteorological echoes were removed during the calculation and an attenuation correction was applied, using polarimetric variables. Due to the KDP estimation technique applied and the bright-band height, the validated RQPE area is within a radius of 10 to 150 km from the center of each radar, so that the bright-band effect is not present. Preliminary validation of the methodology showed a good fit with

surface accumulated precipitation observations (Cancelada et al. 2024). However, some caution is required when analyzing the RQPE values west of RMA1 due to topographic beam blockage. In addition, it is important to notice that major errors and uncertainties are associated with radar beam extinction (specially during wet radome times) and distance from the radar position, leading to precipitation underestimations.

For the object-based verification the first step was to interpolate the ensemble forecasts of each initialization time and the corresponding observations to a common regular grid of 0.025° . Then, MODE was applied to define specific objects in each field and compare their spatial structure. As the identification of objects is based on a convolutional and thresholding process, the results depend on the choice of a convolution radius (R) and a convolution threshold (TH) (see Fig. 29 from Bullock et al. 2016). The greater R is, the greater the smoothing applied to the original fields, resulting in objects with smoother shapes. TH is used to define a mask to create the objects.

In this work, combinations of five values of R (3, 5, 8, 12, and 15 grid lengths, roughly corresponding to 6, 10, 16, 24 and 30 km respectively) and eight values of TH (1, 5, 10, 15, 20, 25, 30 and 45 mm/6 h) were used. This means that, for example, for $R=3$ and $TH=1$ mm/6 h, we were looking for contiguous precipitation areas with a smoothing of 6 km where the 6-h accumulated precipitation is equal to or greater than 1 mm.

3 Analysis verification

3.1 Verification against AWS and radar observations

The observations from AWS were used for monitoring the WRF-LETKF and the SCALE-LETKF systems during the assimilation period. Time series of RMSE and bias, considering observation-minus-background and observation-minus-analysis, averaged over the domain are shown in Figs. 4 and 5 for the SVP case and the SSD case, respectively. A 10-min interval is used for clarification of the time series, as the results for the 5-min interval lead to similar conclusions.

For the SVP case (Fig. 4), the behavior of both WRF and SCALE systems is similar in terms of RMSE for all variables (except for surface pressure in SCALE): a generalized decrease in analysis error is documented concerning background error. This sawtooth pattern indicates a regular forecast error growth together with its reduction during the DA step, as previously documented (eg. Wheatley et al. 2015). For temperature and relative humidity (c.f., Fig. 4a and b,

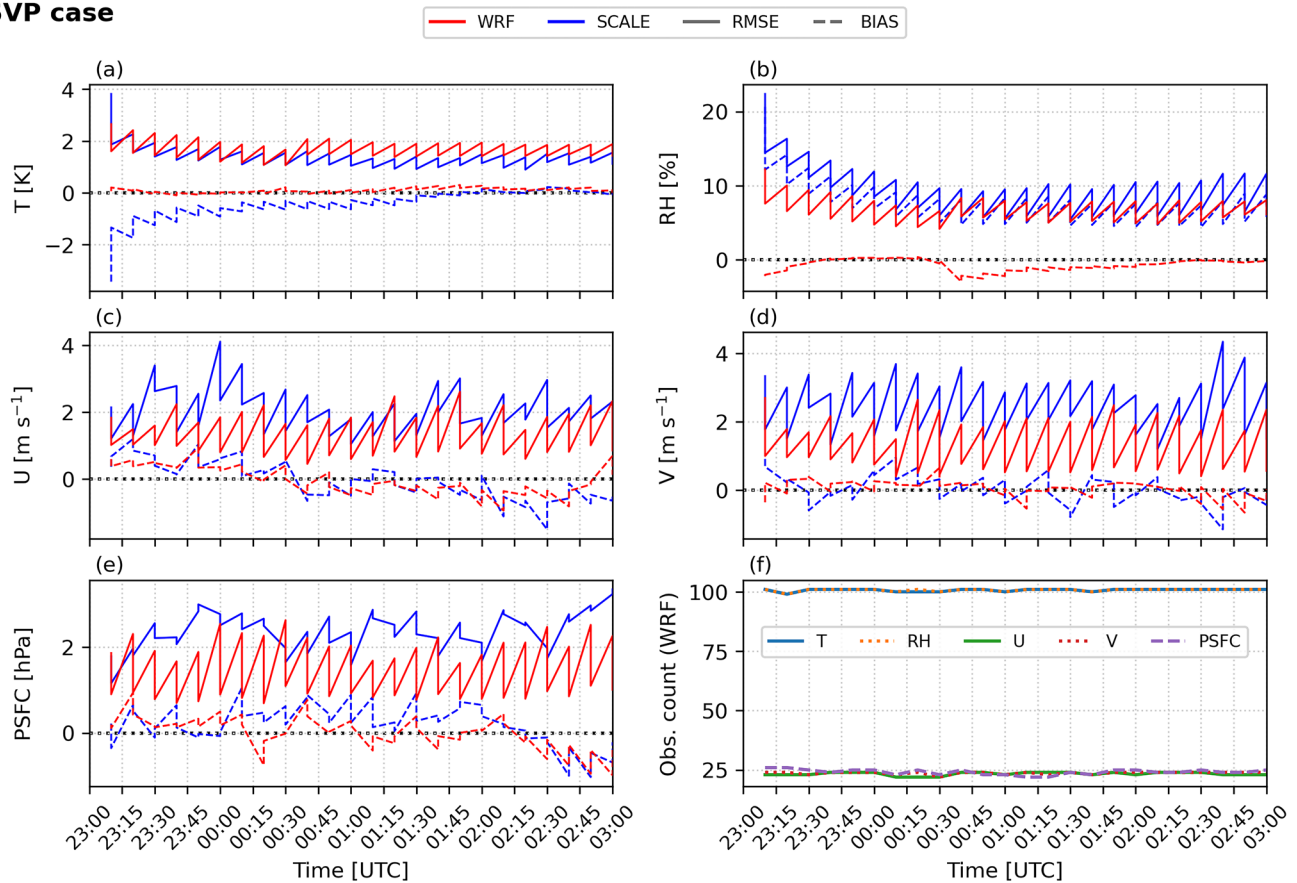
SVP case

Fig. 4 Averaged RMSE (solid line) and bias (dashed line) time series for the SVP case (from 23 UTC 13th to 03 UTC 14th December 2018) considering a 10-min interval observation-minus-background and observation-minus-analysis, for WRF (red) and SCALE (blue)

respectively), a near-zero bias is encountered for WRF, while SCALE remains 10% less humid and 1–2 degrees hotter than the observations during the whole period and the first half of the cycling, respectively. Particularly, it was found that SCALE had a warmer initial state than WRF over the full domain, which was successfully corrected throughout the DA cycling (not shown). The drier condition of SCALE with respect to WRF in this case, was also encountered for the entire domain. These behaviours may be explained by the different land treatment of the systems which directly influences near-surface atmospheric variables (Sect. “[Numerical weather prediction models](#)”): SCALE uses a bucket model, with no explicit vegetation or soil thermodynamics, while WRF uses the more sophisticated Noah-MP, which represent better the land processes. In addition, an increase in the surface pressure bias (less than 1 hPa) is observed after 02 UTC in both models (Fig. 4e). This increase is likely related to differences in the location of the developing convective systems between the analysis mean and the observations, which may produce a spatial displacement of the associated surface pressure features, consequently increasing the

LETKF systems compared to AWS data, considering all the stations. **a** Temperature, **b** relative humidity, **c** zonal wind component, **d** meridional wind component, and **e** surface pressure. The amount of assimilated observations for each variable in the WRF model is depicted in **f**

pressure bias. Finally, Fig. 4f shows the number of observations assimilated for each variable from AWS only for WRF, as the amount for SCALE was almost the same, differing by one or two observations for some cycles. The amount of T and RH assimilated observations is close to 100, four times the amount of wind and surface pressure data, due to the different types of AWS instrument availability.

However, regarding the SSD case, the amount of temperature and relative humidity assimilated observations is close to 60, two times the amount of wind and surface pressure data (c.f., Fig. 5f). This responds to the lower quantity of AWS available in the SSD domain in comparison to the SVP domain (c.f., Fig. 1). For T and RH, WRF generally shows greater values of RMSE than SCALE, particularly for the first half of the cycling period, converging to 2 K and 10% approximately at the end of the simulation (Figs. 5a, b). In addition, WRF has a cold and wet bias in this case (with values decreasing along the cycles), whereas SCALE exhibits dry bias not only for SSD but also for SVP (Fig. 4b). Regarding the RMSE for the wind components and surface

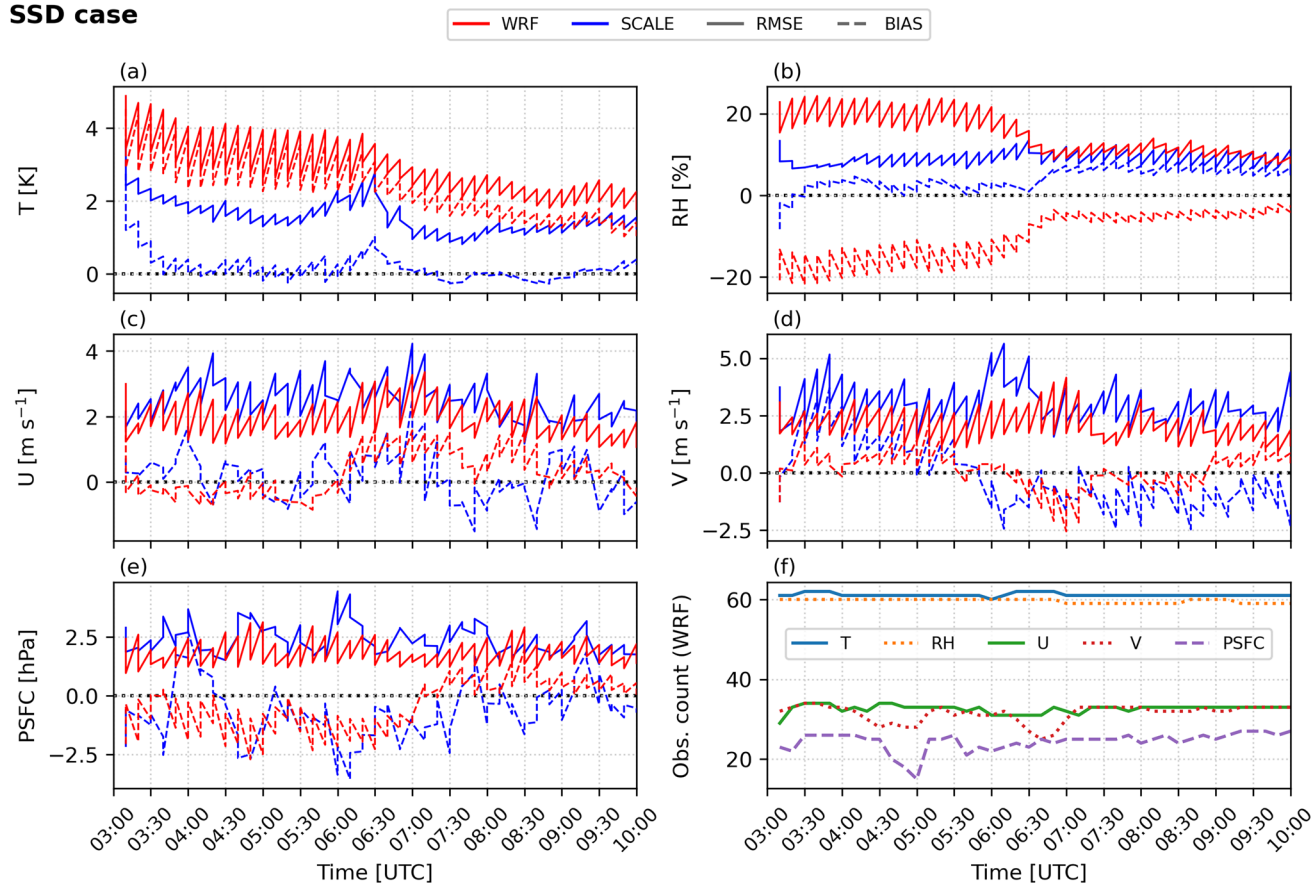
SSD case

Fig. 5 As in Fig. 4, but for the SSD case from 03–10 UTC on October 11th, 2019

pressure, the behavior of both systems is similar, with WRF values generally closer to zero than SCALE (Fig. 5c, d, e).

Equivalent to Figs. 4 and 5, but using weather radar data, Figs. 6 and 7 monitor the performance of both WRF-LETKF and SCALE-LETKF in terms of RMSE and bias of the analysis and background states for the SVP case and the SSD case, respectively.

For the SVP case (c.f., Fig. 6a, b), both models converge to a 10 dBZ error by the end of the DA periods, while WRF shows a larger bias than SCALE, denoting an overestimation of the observed reflectivity. This could be related to the greater number of observations assimilated by WRF (Fig. 6c) which may enhance the convective activity, leading to higher simulated reflectivity values. Moreover, the WRF model exhibits approximately a 1-h spin-up period for reflectivity, with decreasing RMSE and tending to zero bias, while SCALE does not seem to present this issue. For Doppler velocity, WRF shows a smaller RMSE than SCALE, while both tend to have a negative bias towards the end of the DA cycling, when more observations were assimilated (Fig. 6d).

For the SSD case (c.f., Fig. 7a, b), the WRF model presents smaller RMSE values than SCALE, both for reflectivity

and Doppler velocity. However, the bias behaves differently for each variable. For Doppler velocity, both models do not seem to have significant systematic errors, while for reflectivity the bias is large during almost all of the DA period. Additionally, the 1-h spin-up period observed for WRF reflectivity is also present, with a slight increase in RMSE and a negative bias. By 04 UTC, both models start to converge to the observations, but from 06 UTC onwards, the models present opposite bias (i.e., WRF overestimates and SCALE underestimates the observed reflectivity) and a slight increase in RMSE, possibly related to the assimilation of strongly attenuated data from RMA2-Ezeiza radar that was not successfully removed by the quality control process. This is a relevant issue that should be addressed when considering an implementation: a flag could be included in the data when strong attenuation is detected to prevent the assimilation of those observations.

Regarding the number of assimilated observations in both cases (Figs. 6c, d, 7c, d), the models behave similarly for Doppler velocity while differences arise for reflectivity. The SCALE model assimilates two to three times fewer reflectivity observations than WRF given that SCALE limits the amount of observations that can impact a given gridpoint

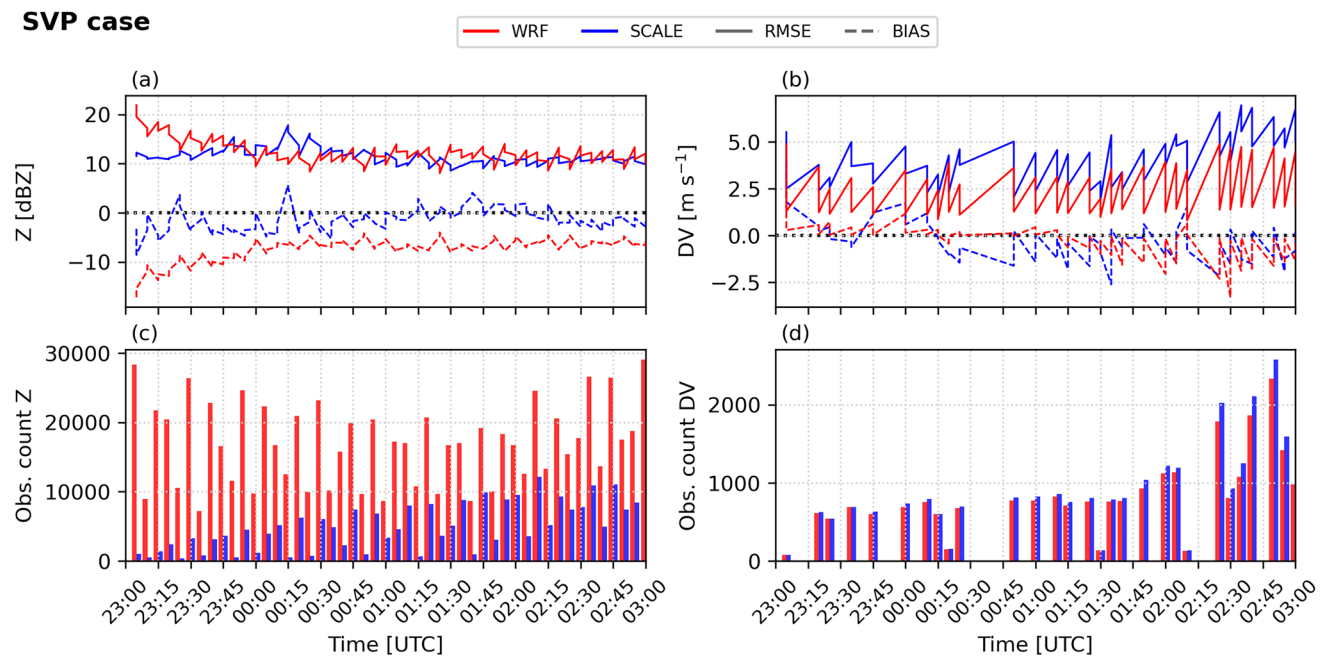


Fig. 6 Averaged RMSE (solid line) and bias (dashed line) time series for the SVP case (from 23 UTC 13th to 03 UTC 14th December 2018) considering a 10-min interval observation-minus-background and observation-minus-analysis, for WRF (red) and SCALE (blue)

LETKF systems compared to weather radar data. **a** Reflectivity (Z), and **b** Doppler velocity (DV). The amount of assimilated observations is depicted for **c** reflectivity and **d** Doppler velocity

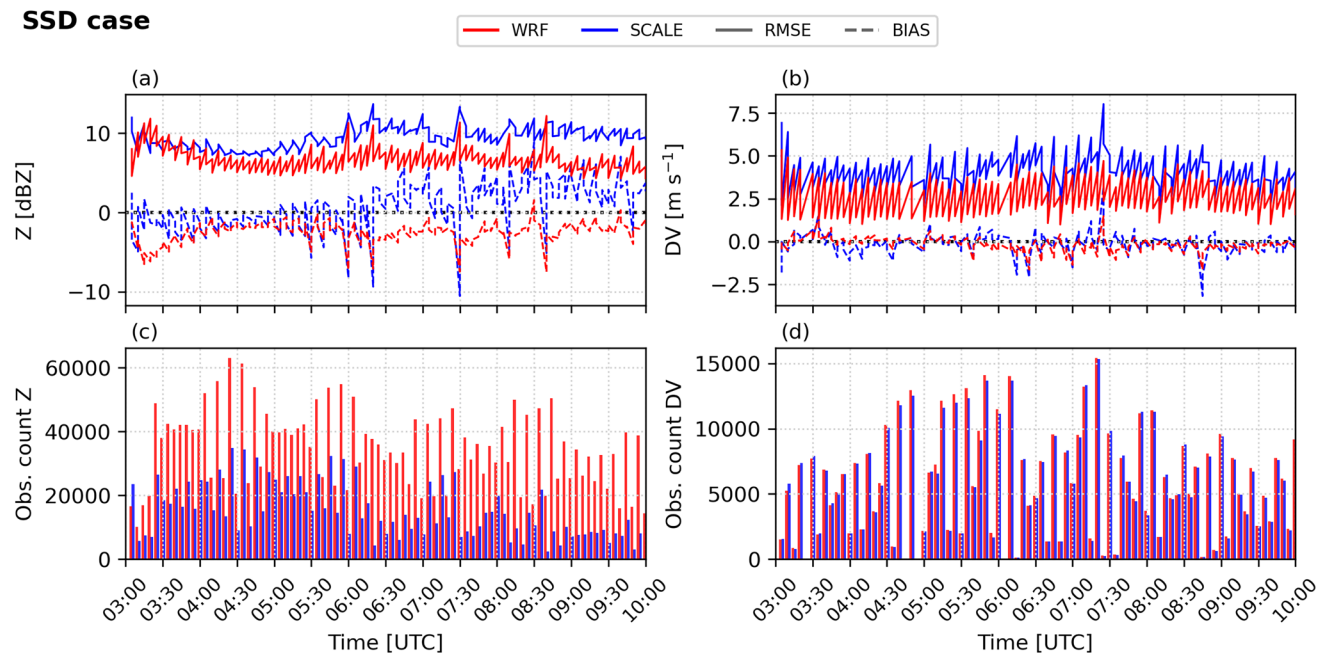


Fig. 7 As in Fig. 6, but for the SSD case from 03–10 UTC on October 11th, 2019

while WRF does not. This difference may lead to a distinct development of the convective systems between the models. Additionally, Doppler velocity data shows a lower refresh rate than reflectivity due to the C-band radar observation strategy, which alternate between two different maximum ranges every 5 min. The highest nyquist velocity is only

acquired while scanning with the shorter maximum range (i.e., every 10 min), leading to the doppler velocity gaps documented (Figs. 6d, 7d). In particular, this feature is more visible for the SVP case, as only one radar is assimilated (c.f., Fig. 1).

3.2 Verification against radiosonde observations

As independent observations of the 3-D atmospheric state variables, the sounding data from RELAMPAGO within the SVP case domain and between 00 and 03 UTC on Dec. 14th, 2018 consisting of 24 soundings was utilized (Schumacher et al. 2021; c.f., Fig. 1). This dataset was not assimilated in the present experiments (Sect. “Assimilated observations”). Figure 8 shows the difference between the observation and the analysis ensemble mean (bias) in zonal wind, meridional wind, temperature, and relative humidity, together with the corresponding RMSE averaged over all 24 soundings. For comparison, the NoDA ensemble forecasts using both models are also depicted.

Both SCALE-LETKF and WRF-LETKF analyses show a reduction of biases in temperature and zonal wind in the lower levels compared to the NoDA (Fig. 8e, g). This is also true for humidity and meridional wind but only for the lowest level (Fig. 8f, h). The RMSE values of the DA systems are equal to or lower than the downscaled forecasts for the levels near the surface (Fig. 8a–d). As expected, the LETKF systems also show a significant reduction of ensemble spread in all variables below 3000 m against NoDA (not shown). This improvement, mostly confined to the lower layers, is consistent with the surface observations assimilation and their impact through the vertical localization scale and planetary boundary layer dynamics. The model analyses still show differences in the biases and RMSE in all the variables from downscaled forecasts in upper levels, although the impacts are not always desirable. For example, an increase in bias is encountered for relative humidity for

the analyses with respect to NoDA experiments, mainly for WRF-LETKF (Fig. 8h). This may be related to the development of the convective systems, which directly affects the column humidity content and if the location is displaced, the metric is degraded. Also, an increase in the RMSE is found for WRF-LETKF for some levels for the 4 variables and SCALE-LETKF for U and V in upper levels, compared to NoDA simulations (Fig. 8a–d). Particularly for SCALE, this does not seem to be associated with a bias in the model since NoDA shows lower values than SCALE-LETKF at those levels (Fig. 8e, f). An encouraging result is that data assimilation reduces temperature bias for SCALE over the whole profile.

4 Forecast verification

In this section the ensemble forecasts initialized from the WRF-LETKF and SCALE-LETKF DA systems are verified. The RQPE calculated for RMA1 and RMA2 (Sect. “Metrics used for verification”) was considered as a basis for the evaluation of simulated precipitation for SVP and SSD cases, respectively. Therefore, all the figures and metrics presented are constructed accounting for RQPE valid area, i.e. between 10 and 150 km from each radar location.

4.1 Grid point forecast error statistics

Figure 9a shows the 6-hourly accumulated RQPE valid at 06 UTC on 14th December 2018 (SVP case), whereas Fig. 9b and c show the ensemble mean forecast precipitation

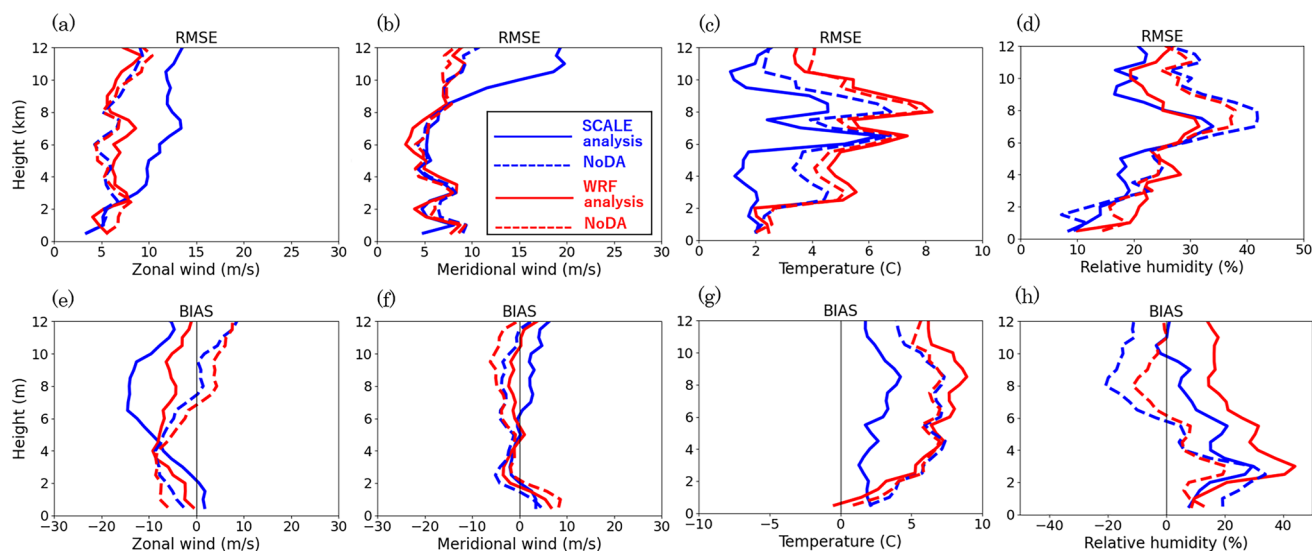


Fig. 8 Averaged vertical profiles of the RMSE (a–d) and bias (considering observation minus analysis) (e–h), calculated from 24 soundings between 00 and 03 UTC of 14th Dec 2018 for the SVP case. **a, e** Zonal wind, **b, f** meridional wind, **c, g** temperature, and **d, h** relative humid-

ity; for WRF-LETKF (red solid line), WRF NoDA (red dashed line), SCALE-LETKF (blue solid line) and SCALE NoDA (blue dashed line). The ensemble mean is considered for all the cases

by WRF and SCALE initialized at 00 UTC, over the same period. In addition, the 20 mm contours from the 4 km WRF ensemble mean (system used as boundary conditions, Fig. 3), is depicted in Fig. 9a to complement the analysis. In this case, both models reproduce the overall distribution of precipitation in the observed area and improve its representation with respect to the coarser available regional forecast. The local peak over 30 mm in the RQPE located near the center of the radar is better captured by the WRF ensemble mean although it is shifted towards the east. Ensemble spread values generally remain low, less than 4 mm/6 h, for both systems (Fig. 9b, c).

Relative frequency histograms were constructed aggregating the 20 ensemble members of each forecast system into a single sample, for comparison purposes (Fig. 9d). The behaviour of WRF, SCALE and 4 km WRF is similar: the greater frequencies are correctly located for the smallest accumulated precipitation amounts as the estimation. However, the models overestimate the frequencies with respect to RQPE for the lower thresholds (10 mm/6 h), particularly

WRF, while they show underestimation for 20, 30 and 40 mm/6 h thresholds.

In terms of probability, both forecasts show at least 10% for 6-h accumulated precipitation greater than 10 mm over almost all the regions where it effectively happened (Fig. 9e, f). Specifically, WRF probability for this threshold reaches 90% (Fig. 9e) while SCALE probability reaches 60% (Fig. 9f). In addition, considering a 20 mm threshold WRF (SCALE) showed probability values of 60% (40%) (not shown).

The 6-hourly accumulated RQPE valid at 12 UTC on 11th October 2019 (SSD case), together with the ensemble mean forecast precipitation by WRF and SCALE initialized at 06 UTC, over the same period, are shown in Fig. 10a–c. The ensemble mean accumulated precipitation from both models shows different patterns compared to the RQPE. The WRF forecast has, to some extent, a better agreement with the observation in the location of the local peak, with the maximum slightly shifted to the southeast. While SCALE tends to produce the maximum rainfall to the north from the precipitation peak observed by the radar. Considering the

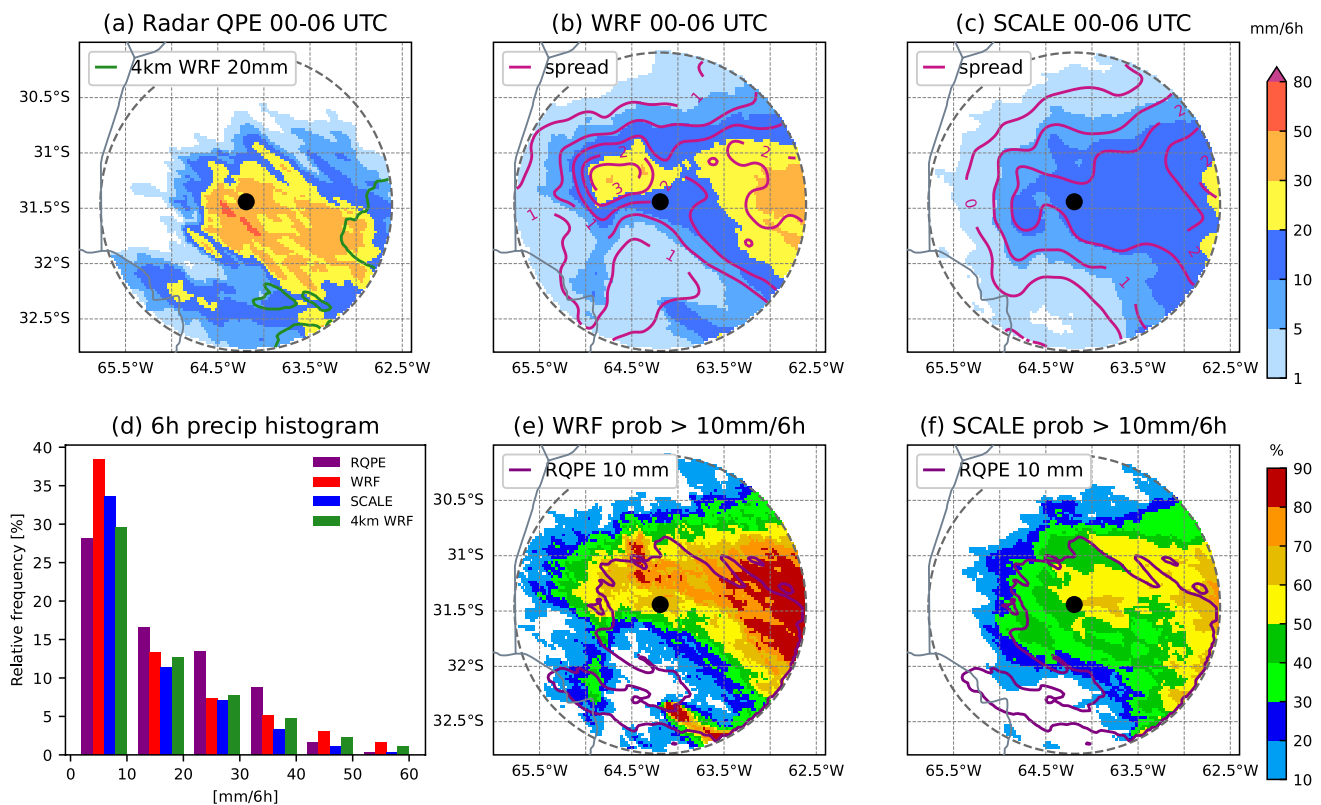


Fig. 9 6-h accumulated precipitation valid at 06 UTC 14th December 2018 (SVP case) for **a** RQPE, **b** WRF forecast initialized at 00 UTC, **c** SCALE forecast initialized at 00 UTC. The green contours in **a** indicate 4 km WRF SAP.SMN 20 mm ensemble mean forecast; color shades and contours in **b–c** indicate ensemble mean and spread, respectively. **d** Frequency histograms for RQPE, WRF, SCALE and 4 km WRF, considering each 20 ensemble members. Probability (shaded)

of 6-h accumulated precipitation over 10 mm valid at 06 UTC 14th December 2018 for **e** WRF forecast initialized at 00 UTC, **f** SCALE forecast initialized at 00 UTC; the RQPE 10 mm contours are indicated with purple. All the values are masked in the area closer than 10 km or further than 150 km from the radar, according to the RQPE calculation

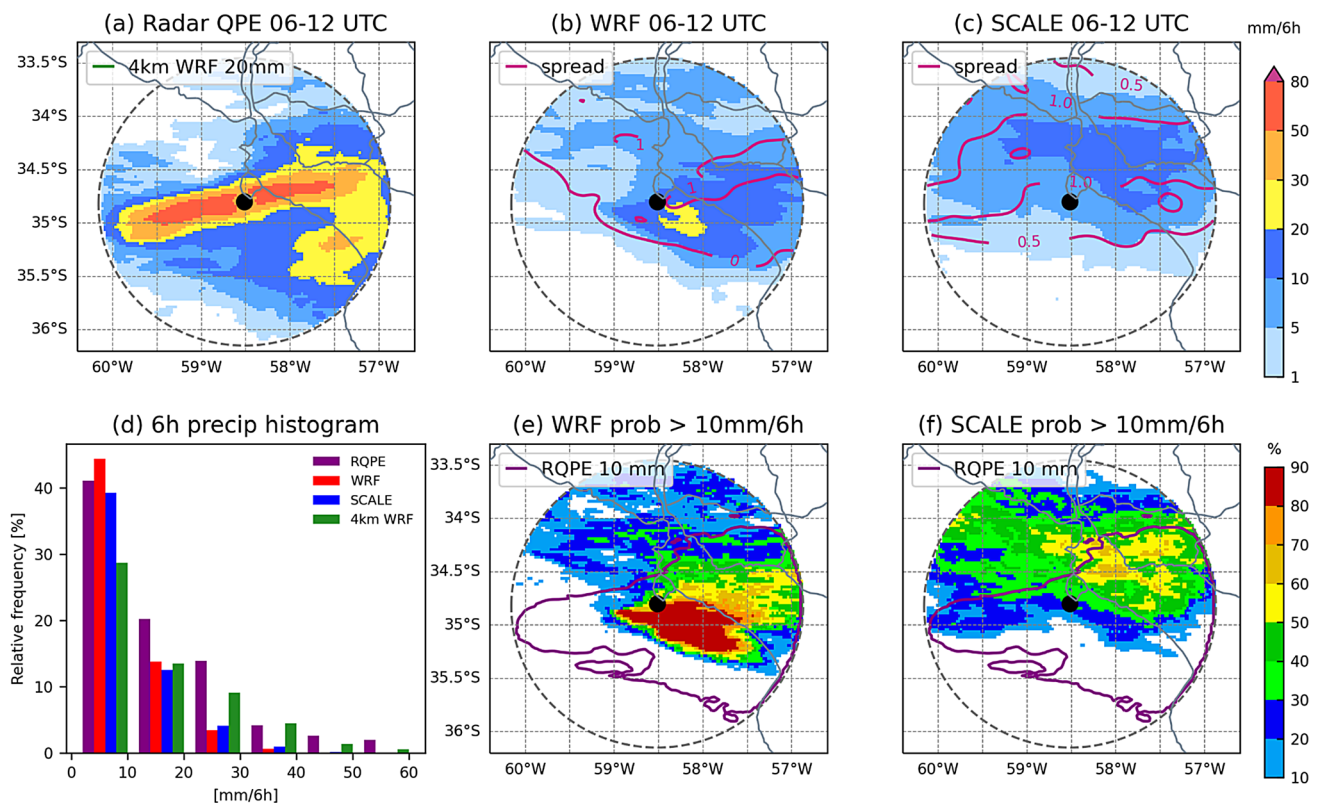


Fig. 10 6-h accumulated precipitation valid at 12 UTC 11th October 2019 (SSD case) for **a** RQPE, **b** WRF forecast initialized at 06 UTC, **c** SCALE forecast initialized at 06 UTC. The green contours in **a** indicate 4 km WRF SAP.SMN 20 mm ensemble mean forecast; color shades and contours in **b**, **c** indicate ensemble mean and spread, respectively. **d** Frequency histograms for RQPE, WRF, SCALE and 4 km WRF,

considering each 20 ensemble members. Probability (shaded) of 6-h accumulated precipitation over 10 mm valid at 12 UTC 11th October 2019 for **e** WRF forecast initialized at 06 UTC, **f** SCALE forecast initialized at 06 UTC; the RQPE 10 mm contours are indicated with purple. All the values are masked in the area closer than 10 km or further than 150 km from the radar, according to the RQPE calculation

ensemble mean, both high resolution DA systems show an improvement in the location of the forecasted rainfall with respect to the one from the 4 km WRF, which in fact did not represent areas with 20 mm/6 h inside the valid RQPE region (Fig. 10a). Equally to the SVP case, ensemble spread values remain low, less than 2 mm/6 h in this case for both systems (Fig. 10b, c). Although the ensemble means underestimate the accumulated RQPE, the highest probability values of exceeding 10 mm in 6-h reached by WRF and SCALE inside the area where effectively precipitated that amount were 90 and 60%, respectively (Fig. 10e, f). Regarding the frequency histograms, the results for the SSD case are similar to the SVP one, but in this case only the WRF ensemble overestimates the lowest accumulated precipitation amount frequency (Fig. 10d). Moreover, for thresholds greater than 20 mm/6 h the 4 km WRF ensemble presents the closer frequency to the estimation.

In addition, considering the hydrological purposes of these systems (Sect. “Introduction”), it is of interest analyzing higher frequency precipitation. Therefore, specific valid times are chosen for each case that match with the corresponding maximum amounts of precipitation. For the SVP

case, the hourly accumulated RQPE indicates a peak rainfall exceeding 30 mm at 04 UTC (Fig. 11a). Ensemble mean forecasts initialized at 02 UTC are presented in Fig. 11b and c. WRF forecast ensemble mean shows a narrower structure than that of SCALE, with higher amount of precipitation, being a more coherent structure closer to the observed one. In comparison with the 5 mm contours from the 4 km WRF ensemble mean, it is shown that both models also improve the representation of the one hour accumulated precipitation (Fig. 11a, b, c). Although the system reproduced by WRF is shifted to the northeast of the observed one, the probability of precipitation greater than 10 mm in 1-h reaches values of 90%, representing valuable information for the forecasts (Fig. 11e). In this case, the probability values for SCALE are lower than 30% (Fig. 11f).

To complement these results, the pFSS (Sect. “Metrics used for verification”) of hourly precipitation, considering RQPE as the verifying truth, were calculated considering a neighborhood of 10 km and two threshold values. The forecasts initialized at 00 UTC on 14th December 2018 (SVP case) were used for the pFSS depicted in Fig. 11d, as functions of forecast valid times. Results for the 4 km WRF and

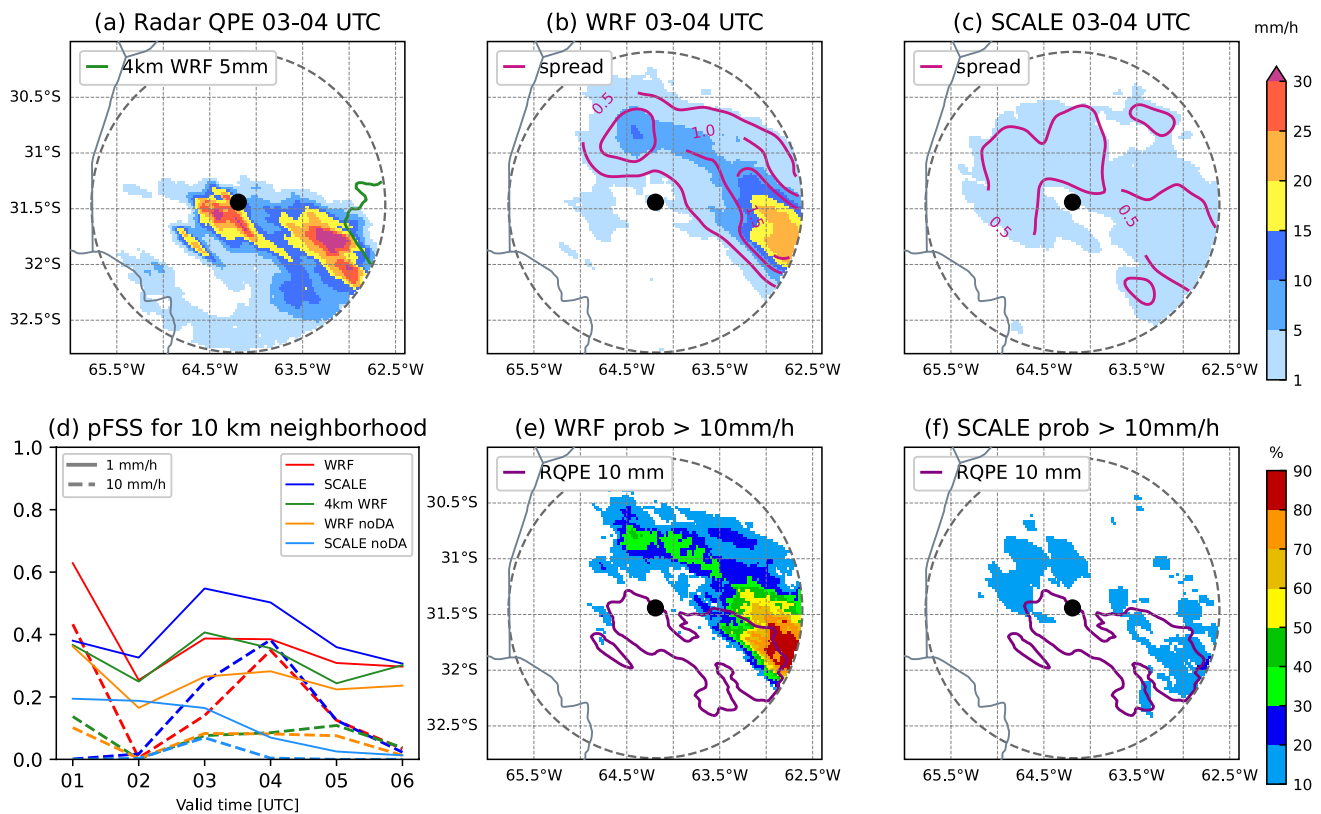


Fig. 11 Hourly accumulated precipitation at 04 UTC 14th December 2018 (SVP case) for **a** RQPE, **b** WRF forecast initialized at 02 UTC and **c** SCALE forecast initialized at 02 UTC. The green contours in **a** indicate 4 km WRF SAP.SMN 5 mm ensemble mean forecast; color shades and contours in **b**, **c** indicate ensemble mean and spread, respectively. **d** Probabilistic Fractions Skill Scores of WRF, SCALE, 4 km WRF, WRF noDA and SCALE noDA ensemble forecasts for the thresholds of 1 and 10 mm accumulated precipitation in

1 h, for a 10 km neighborhood, as functions of the valid time for the 00 UTC 14th December initialization, with respect to RQPE (expected value = 1). Probability (shaded) of 1-h accumulated precipitation over 10 mm valid at 04 UTC 14th December 2018 for **e** WRF and **f** SCALE forecasts initialized at 02 UTC; the RQPE 10 mm contours are indicated with purple. All the values are masked in the area closer than 10 km or further than 150 km from the radar, according to the RQPE calculation

NoDA ensembles are also included. Overall, both models show similar performances in this case for each threshold, scoring better than NoDA systems. For the first hour, WRF ensemble outperforms the other systems for both thresholds, but the improvement due to DA decays rapidly. In particular, for 1 mm/h SCALE shows the best performance after the second forecasted hour, while WRF performance remains near 4 km WRF. The behaviour of SCALE and WRF outperforming their corresponding NoDA systems within pFSS metric is consistent through different neighborhoods and thresholds, while 4 km WRF sometimes show greater values of pFSS for specific combinations (not shown).

For the SSD case, the hourly accumulated RQPE indicates a peak rainfall between 15 and 25 mm at 07 UTC (Fig. 12a). Both WRF and SCALE ensemble mean forecasts show precipitation areas in the domain but in different locations, with underestimated values, for the 3-h forecast lead time valid at 07 UTC (Fig. 12b, c). The WRF forecast shows a broad area of precipitation near the observed location, whereas the SCALE forecast has more intense precipitation

in the northern part of the domain. In addition, the representation of the 5 mm contours by these systems seem better than that of the 4 km WRF SAP.SMN forecasts (Fig. 12a, b, c). Regarding the probability of 1-h accumulated precipitation greater than 10 mm, WRF shows values of 40% inside the verifying area and SCALE reaches values of 60% a bit shifted to the north (Fig. 12e, f). Lastly, the pFSS was also calculated with a focus on the period of more intense rain, for the forecasts initialized at 04 UTC on 11th October 2019. The same parameters as the SVP case were used to construct Fig. 12d. Contrary to what we found for the SVP case, the WRF forecast shows higher pFSS than the other systems for almost all the valid times, for the two thresholds. Again, the major DA impact is achieved for the first hour for both high resolution systems. In this case, WRF 4 km generally outperforms SCALE, but this reverts for greater neighborhood lengths (not shown).

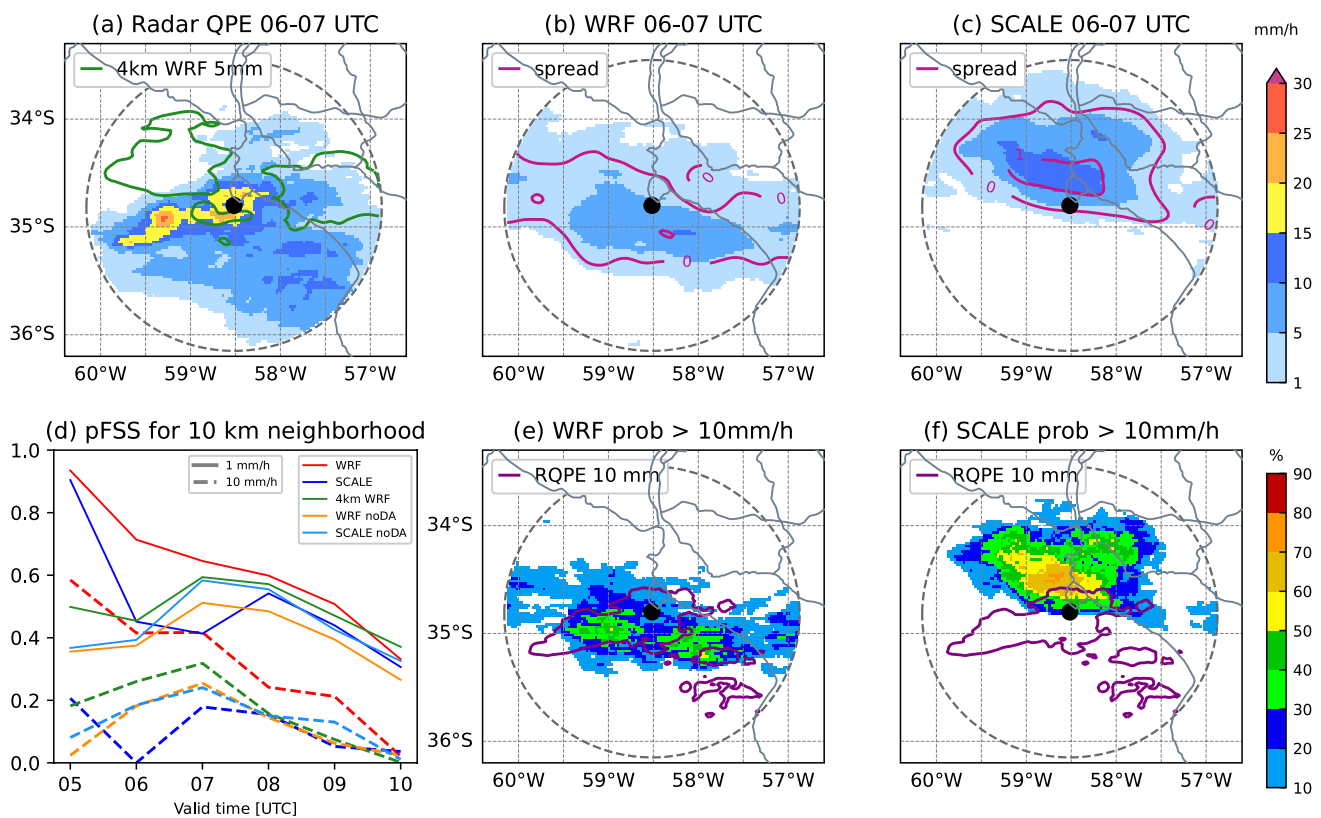


Fig. 12 Hourly accumulated precipitation at 07 UTC on 11th October 2019 (SSD case) for **a** RQPE, **b** WRF forecast initialized at 04 UTC and **c** SCALE forecast initialized at 04 UTC. The green contours in **a** indicate 4 km WRF SAP.SMN 5 mm ensemble mean forecast; color shades and contours in **b-c** indicate ensemble mean and spread, respectively. **d** Probabilistic Fractions Skill Scores of WRF, SCALE, 4 km WRF, WRF noDA and SCALE noDA ensemble forecasts for the thresholds of 1 and 10 mm accumulated precipitation in 1 h, for

a 10 km neighborhood, as functions of the valid time for the 04 UTC 11th October initialization, with respect to RQPE (expected value = 1). Probability (shaded) of 1-h accumulated precipitation over 10 mm valid at 07 UTC 11th October 2019 for **e** WRF and **f** SCALE forecasts initialized at 04 UTC; the RQPE 10 mm contours are indicated with purple. All the values are masked in the area closer than 10 km or further than 150 km from the radar, according to the RQPE calculation

4.2 Object-based forecast verification

The MODE tool was used to perform a feature-based verification of 6-h accumulated precipitation forecasts with respect to RQPE (Sect. “Metrics used for verification”). For the SVP case, the results of the seven forecast initializations for the 20 members of SCALE and WRF were considered (i.e. 140 6-h accumulated precipitation fields were analyzed for each model), while for the SSD case, the quantity of forecast initializations was 13 (i.e. 260 fields were analyzed for each model).

At a first step, taking into account all the combinations of the smoothing radius R and the precipitation thresholds TH (Sect. “Metrics used for verification”), we evaluated different attributes:

The interest value, which summarizes information of different attributes calculated for the matched objects, with a fuzzy logic algorithm that is applied to identify

similarities between forecasted and observed objects (ranges between 0 and 1);

The area ratio, which is defined as the forecast object area divided by the observation object area, providing a measure of whether there is an over- (values greater than 1) or under- (values less than 1) prediction of the areal extent forecasted;

The centroid distance, which is the distance (in grid units) between the centroids of the observed and predicted objects (the smaller, the better).

Figures 13 and 14 show the dependence of these attributes with the thresholds and convolutional radius, for each case study.

For the SVP case, similar numbers for the mean interest values are found for both models (Fig. 13a). For TH up to 30 mm/6 h the maximum values reached are 1 for almost all the R considered, indicating the existence of cases with a good agreement between the observed and forecasted objects. Regarding the area ratio, both models generally

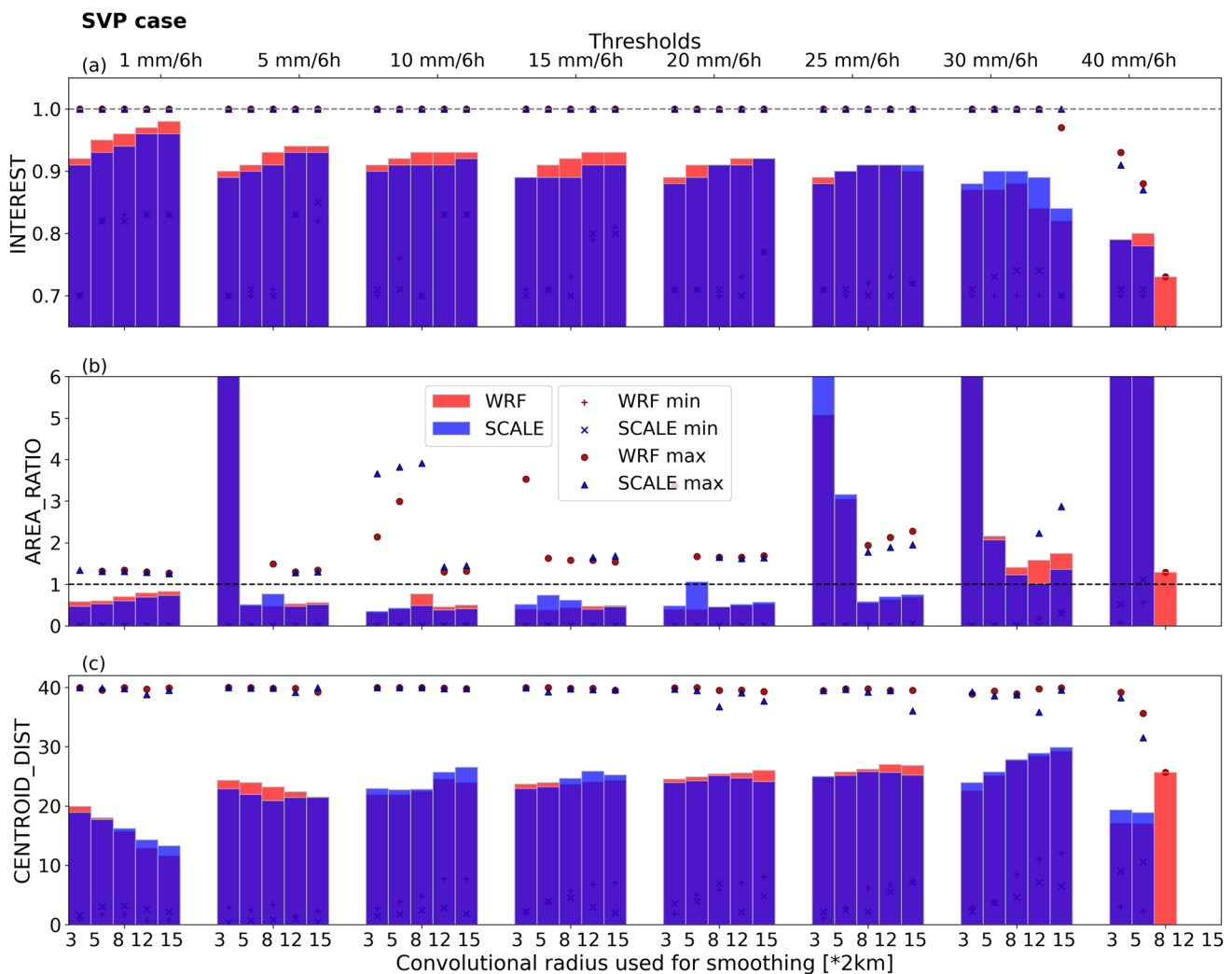


Fig. 13 Minimum, mean, and maximum values considering the 20 ensemble members and all the forecast initializations for the SVP case for the following MODE attributes: **a** interest; **b** area ratio, and **c** centroid distance; for WRF and SCALE simulations. The results

are shown for all the precipitation thresholds indicated in the upper axis (1, 5, 10, 15, 20, 25, 30, and 40 mm/6 h), and the convolutional radius used for smoothing indicated in the lower axis (3, 5, 8, 12 and 15 [*2 km]). In (a and b) the dashed line indicates the expected value 1

show similar values: underestimations are common for TH less than 25 mm/6 h, while overestimations are found when considering R of 3 for some TH, and for R equal to 5 and TH 40 mm/6 h (Fig. 13b). Also, similar mean values of the centroid distance are found for WRF and SCALE (Fig. 13c). It is worth mentioning that in this case for the combinations of TH-R given by 40–12 and 40–15 there were not any identified observed objects, therefore there are not associated matches.

For the SSD case, the area ratios evidence underestimations or even values very close to 1 for both models, except for some combinations of the smallest R (Fig. 14b), while for the centroid distances SCALE results in larger values than WRF for the majority of the thresholds and radii (Fig. 14c). When analyzing the mean values of the interest score, similar numbers are found in both models for TH less

or equal to 15 mm/6 h, while for higher TH WRF shows the closest values to one (Fig. 14a). In this case, for TH up to 25 mm/6 h the maximum values reached of the interest score are 1 for almost all the R considered.

At a second step, the objects defined by TH 10 mm/6 h and R 8 were selected to get insight into the representation of the centroid locations by the models, with the aim of studying possible displacements of the forecasted objects. The selection of these specific values of TH and R responds to the evaluation of all the combinations shown in Figs. 13 and 14, and is considered representative of a general behaviour of the objects. For Fig. 15 all the ensemble members and four (five) forecast initializations for the SVP (SSD) case are used. The observation centroids are depicted along with dashed lines indicating the initial and final positions of the main system for each case: their movement is slow in the

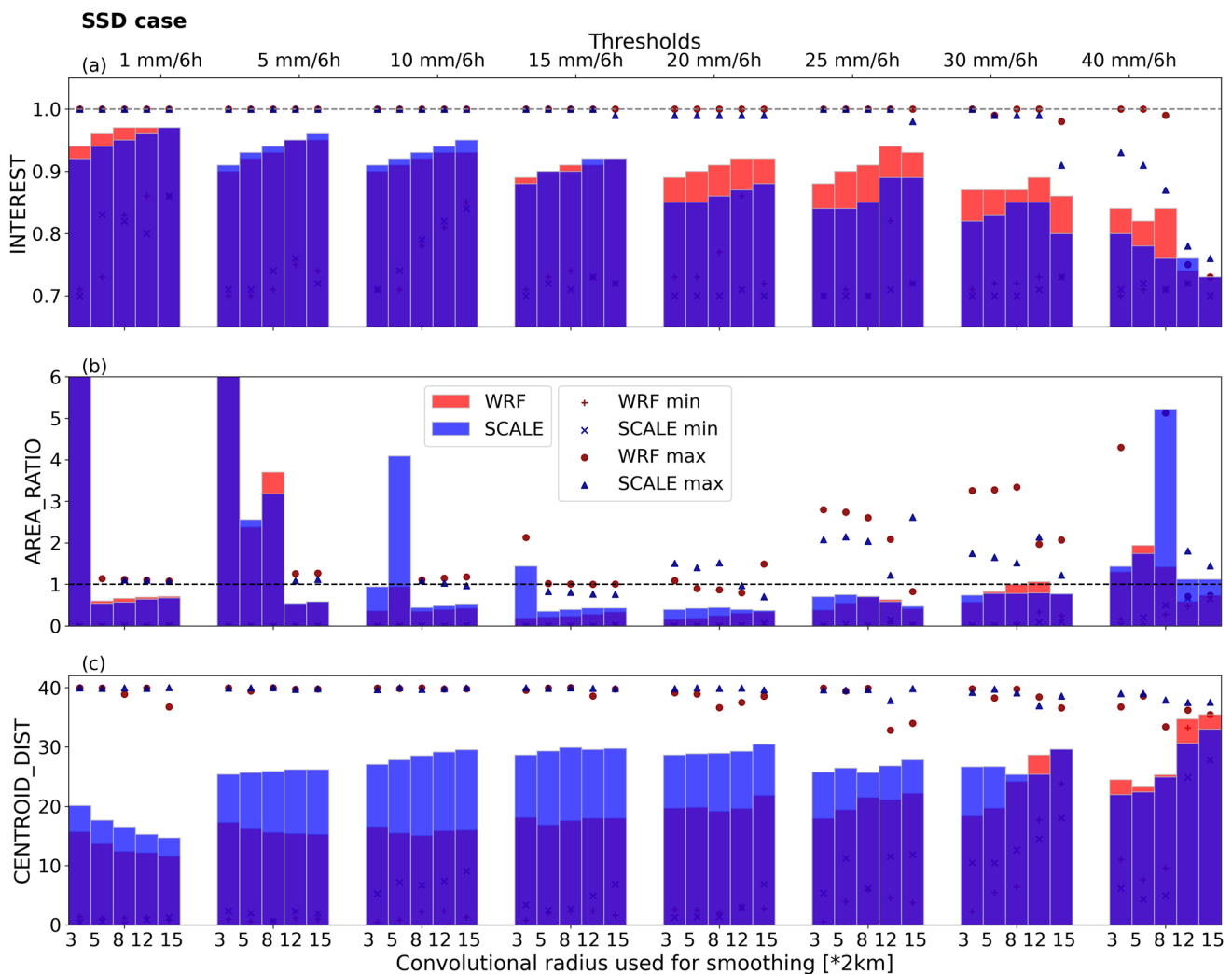


Fig. 14 Same as Fig. 13 but for the SSD case

northeast and north-northeast directions for SVP and SSD cases, respectively. Overall, both models forecasted systems around the ones observed for the SVP case and particularly matched according to MODE interest attributes greater than 0.7 (dark gray markers) (Fig. 15a, b). Regarding the SSD case, the matched objects reveal a general southeast shift for WRF (Fig. 15c) and a north shift for SCALE (Fig. 15d), in concordance with the results shown in Sect. “[Grid point forecast error statistics](#)”. Analyzing the members separately, there is no common behavior for each model for all the members and times considered, nor is there a clear signal of boundary conditions’ influence (not shown).

5 Summary and discussion

We have developed and evaluated the first prototypes of high-resolution and rapid-update data assimilation and numerical weather prediction systems under the PREVENIR

project, and investigated the predictability of two cases of significant rain events that occurred over the target basins in Córdoba (SVP) and Buenos Aires provinces (SSD), Argentina. We utilized two regional numerical weather prediction models, namely WRF and SCALE. Both systems used the LETKF method with 40 ensemble members, 2 km horizontal resolution, and 5-min cycling to assimilate observational data from C-band Doppler weather radars and automatic surface weather stations. Extended 6-h ensemble forecasts with 20 members were initialized every 30 min from the analysis ensemble. The performance of SCALE-LETKF and WRF-LETKF forecast systems were investigated with the aim of evaluating both prototypes rather than comparing the models, and taking into account the feasibility of a subsequent transition from research to operations for the PREVENIR system.

Several limitations of this work should be acknowledged: (a) the selection of only two case studies for the performance evaluation (Sect. “[Pilot basins and test cases](#)”);

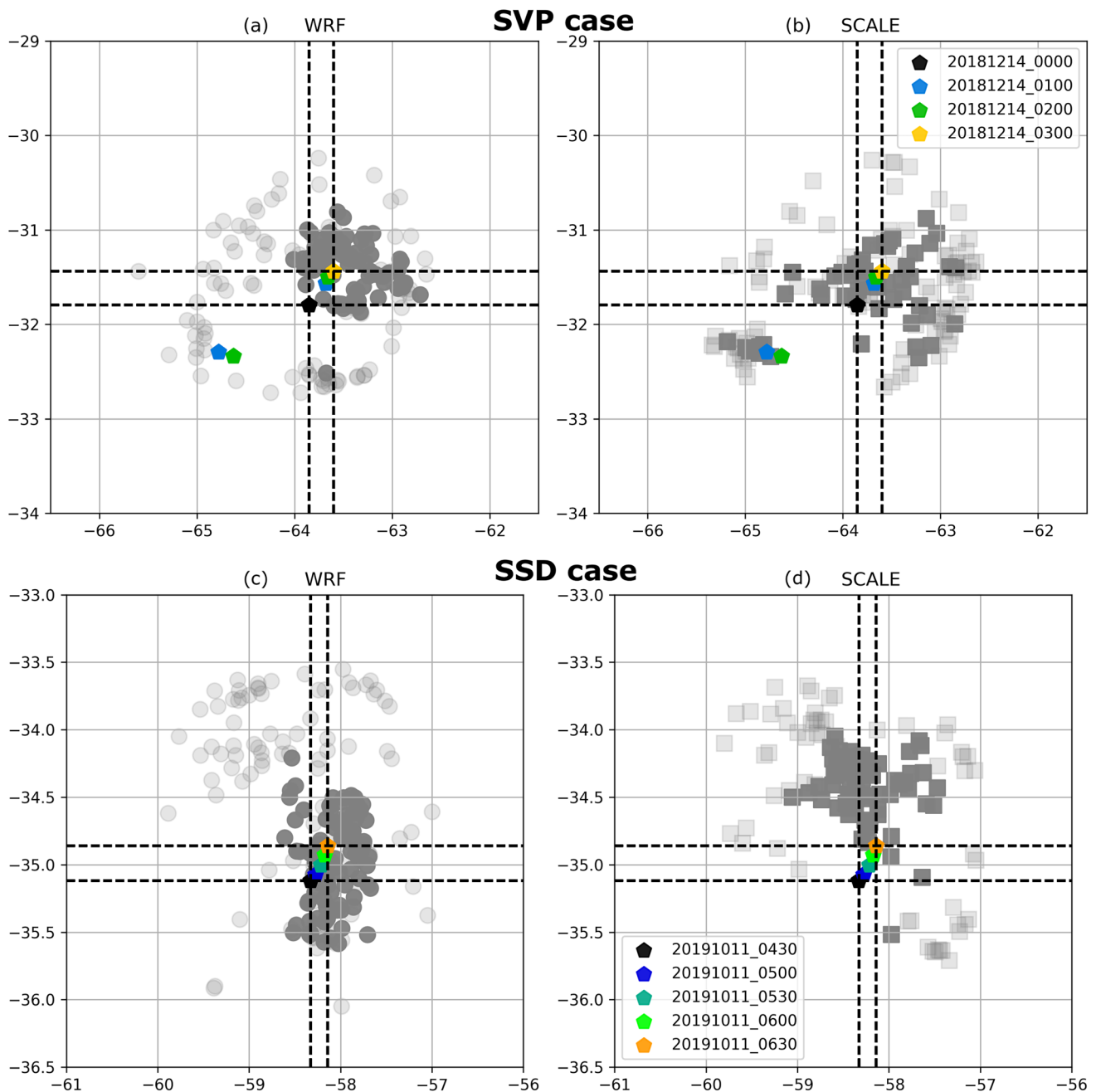


Fig. 15 Centroid locations (latitudes, longitudes) defined by MODE objects for **a, c** WRF and **b, d** SCALE simulations and RQPE (pentagon); for TH 10 mm/6 h and R 8, considering the 20 ensemble members; for **a, b** SVP case and **c, d** SSD case. The different colors of RQPE centroids respond to distinct forecast initializations, indicated in

the legend. In each panel, the dark gray circles/squares correspond to the model centroids that matched the observations with interest greater than 0.7. The dashed lines correspond to the initial and final RQPE centroid location

(b) the differences in the models' configurations, i.e. multi-physics ensemble and Noah-MP land model (WRF) versus a unique physics suit and a bucket land model (SCALE) (Sect. "Numerical weather prediction models"); (c) the differences in reflectivity treatment for DA, leading to a major amount of observations assimilated by WRF-LETKF with respect to SCALE-LETKF (Sects. "Ensemble

data assimilation and forecast system" and "Verification against AWS and radar observations"); (d) the reliance on RQPE as the verifying truth for precipitation, which presents some sources of uncertainties (Sect. "Metrics used for verification").

Despite these limitations, the results suggest that the prototypes could offer several advantages for hydrological

applications compared with the coarser forecasting resources available in the region and NoDA systems. These advantages include more representative probabilistic forecasts of 1-h and 6-h accumulated precipitation (Sect. “[Grid point forecast error statistics](#)”), improved localization of precipitating systems (Sect. “[Forecast verification](#)”), more frequently updated forecasts based on consistent ensemble of analyses (Sect. “[Analysis verification](#)”), and enhanced representation of precipitation at finer spatial scales (Sect. “[Forecast verification](#)”).

It is worth mentioning that these two evaluated case studies gave us insights into the issues we need to further investigate to improve the performances of both WRF-LETKF and SCALE-LETKF systems. The systematic bias found in the surface observation statistics indicates model limitations in the physics schemes representing boundary layer and surface processes. The difference in the initial states of WRF and SCALE at the time of starting the data assimilation, probably affected the impact of the observations throughout the cycling, leading to distinct updates for each LETKF system. Regarding the ensemble spread control, we may need to revisit the LETKF parameters such as the observation error standard deviation and the factor of relaxation to prior perturbation. With respect to the observations, attention should be driven to the treatment of attenuated reflectivity data, to prevent assimilating such detrimental observations. Also, to examine its impact on the precipitation forecast accuracy and find the optimal settings, sensitivity experiments should be performed systematically using a sufficiently long period of data. Finally, a bias correction may be considered to adjust as much as possible the forecasted precipitation to the observed one (eg. Tang et al. 2026).

Moreover, in our case studies, initial and boundary conditions are downscaled from the operational ensemble forecasts of the SAP.SMN, which was running as an ensemble downscaling from the National Centre for Environmental Prediction GFS ensemble without data assimilation. Since December 2024, the Argentine National Meteorological Service has implemented operationally a new forecast system with hourly data assimilation cycling using frequent and local observations in Argentina, including radar reflectivity. In subsequent studies, we plan to assess the impact on the precipitation forecast of the PREVENIR system by using the initial and boundary conditions from the ensemble forecast data from the new operational system with reduced uncertainty in synoptic and mesoscale processes. In addition, there are newly installed radars in the region, which would provide more observations, probably improving the analysis quality in future case studies.

Precipitation forecasts through convective-scale radar data assimilation and high-resolution numerical weather prediction models show potential but are still challenging in

many aspects (Hu et al. 2023; Sun 2025). Generally, it has less accuracy in shorter lead times compared to simpler nowcasting methods. Therefore, it is a common practice to use a hybrid system combining forecasts by numerical weather prediction and nowcasting (Imhoff et al. 2023). Under the PREVENIR project, the update of the nowcasting system using radar observation is underway. The hybridization of the two systems to provide optimal precipitation forecasts in each time scale is one of our future works.

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Data availability The data that support the findings of this study may be accessed upon request.

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